

Sri Lanka Institute of Information Technology



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Project Topic Name : DIET SAFE BOT - Predicting Food Healthiness

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Artificial Intelligence & Machine Learning | IT2110

B.Sc. (Hons) in Information Technology

1. Introduction and Problem Statement

The DIETSAFE BOT is an AI-powered solution for food safety and nutrition, leveraging machine learning to enhance dietary safety and support health-conscious cooking for home cooks, restaurants, and healthcare providers. It provides an AI-driven model for healthy recipe classification, analyzing nutritional data and recipe details to classify recipes as healthy or unhealthy.

In today's fast-paced world, real-world problems such as diet-related health issues (e.g., obesity, diabetes), manual recipe analysis, nut and mushroom allergies in recipes, and limited accessibility for dietary needs pose significant challenges. Nut and mushroom allergies affect 1-2% of the population, with severe allergic reactions (anaphylaxis) requiring accurate identification of safe recipes. Diet-related diseases are rising globally, making it essential to promote wellness through informed meal choices.

The aim of this project is to develop an AI-powered machine learning model to classify recipes from the Epicurious Recipe Dataset based on healthiness, ensuring safe and nutritious dining options. This addresses critical challenges in food safety and promotes wellness through healthy meal choices. The bot analyzes nutritional features such as fat content, calories, and sodium levels to determine recipe healthiness.

The project spans domains like Healthcare (supports allergy management and wellness goals) and Food & Nutrition (ensures safe and healthy recipe recommendations). The objective is to achieve >90% accuracy using a machine learning classification approach, developed collaboratively, while mitigating biases for ethical deployment.

2. Dataset Description

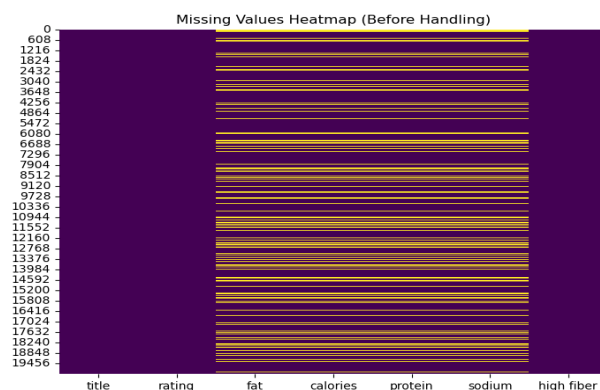
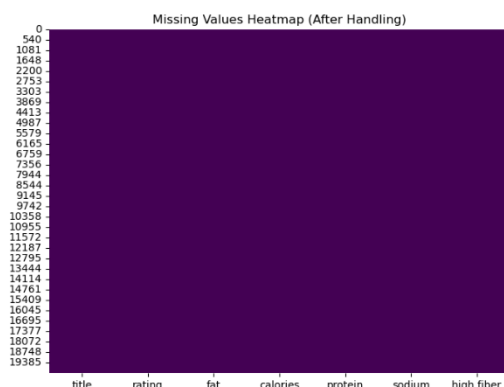
- **Source:** Kaggle (Epicurious - Recipes with Rating and Nutrition dataset, scraped from Epicurious.com by Hugo Darwood)
- **Size:** 20,052 records (rows), 680 features (columns)
- **Target Variable:** No direct target in the raw dataset; derived as binary "healthy" (1 if `health_score > median ~0.5` during feature engineering; approximately 5,927 healthy, 4,692 unhealthy in processed version). This derivation balances positive (e.g., high protein/fiber) and negative (e.g., high fat/sodium) nutritional aspects.
- **Class Imbalance:** Mild skew (roughly 56:44 in processed data), addressed with SMOTE (Synthetic Minority Over-sampling Technique) to balance classes during model training.
- **Key Features:**
 - Numerical: rating (0-5 scale), calories, protein, fat, sodium (continuous values, with some extremes like calories up to millions due to data errors).
 - Text: title (recipe names, e.g., "Lentil, Apple, and Turkey Wrap").
 - Categorical/Binary: 674 binary flags (0/1) for recipe tags/categories, such as #cakeweek, #wasteless, 22-minute meals, 3-ingredient recipes, alcoholic, almond, appetizer, apple, etc. These represent ingredients, occasions, dietary preferences, and preparation methods (e.g., bake, boil, brunch).
- **Limitations:** Western-centric bias (recipes primarily from Epicurious, focusing on American/European cuisines, potentially underrepresenting global diversity); significant missing values in nutritional columns (e.g., ~20% in calories, protein, fat, sodium); some erroneous extreme values (e.g., impossibly high calories); binary tags are sparse (most are 0 for any given recipe). No timestamps, user demographics, or ingredient lists beyond tags, limiting deeper analysis.
- **Relevance:** Nutritional features (calories, fat, etc.) directly correlate with health metrics, while tags enable category-based filtering. The dataset is ideal for deriving health predictions, with EDA revealing skewed distributions (e.g., right-skewed calories) and opportunities for feature engineering (e.g., binning fat levels).

3. Preprocessing & EDA

Preprocessing was collaborative, focusing on data quality for BERT. Steps: handling missing data, encoding, outlier removal, feature engineering, selection, scaling. EDA (histograms, heatmaps, boxplots) justified decisions.

3.1 Handling Missing Data (IT24102307_Handling_Missing_Data.ipynb)

- **Technique:** Selected columns (title, rating, fat, calories, protein, sodium, high_fiber); dropped rows with missing values; removed duplicates on 'title'; mean-imputed 'rating'.
- **Justification:** Ensures completeness; high missing rates (e.g., 4,183 in fat) avoided bias.
- **Impact:** Reduced to 10,620 rows.
- **EDA:** Heatmaps confirmed no post-handling misses.

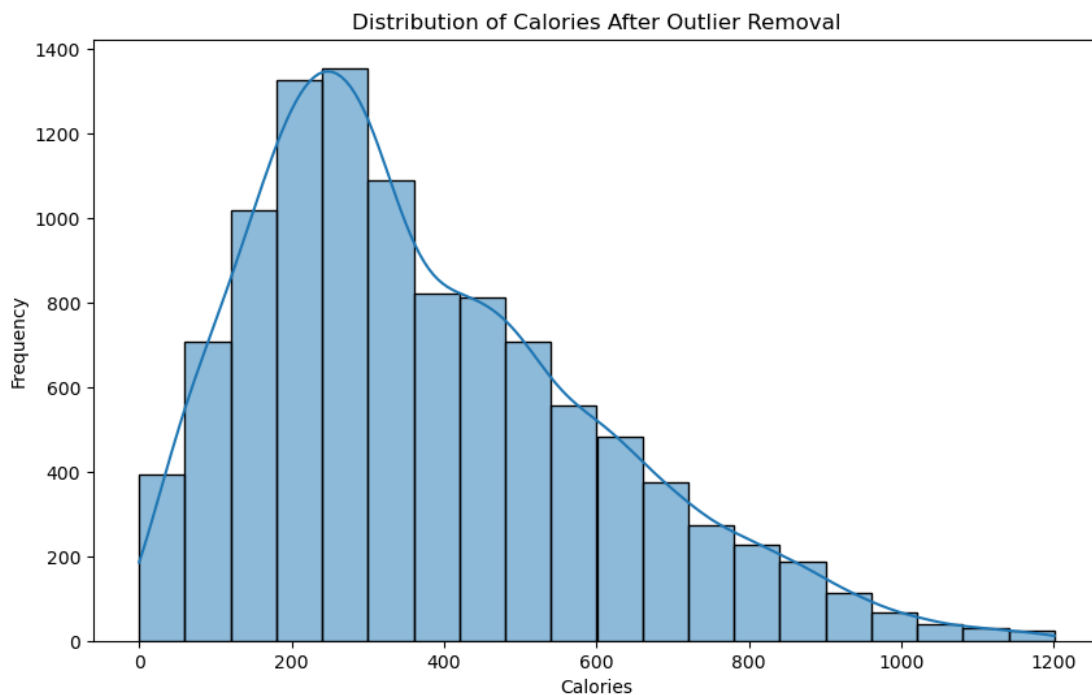


3.2 Encoding Categorical Data (IT24102241_Encoding_Categorical.ipynb)

- **Technique:** Binned 'fat' (low/medium/high); one-hot encoded (fat_level_medium/high). High_fiber binary.
- **Justification:** Numerical for ML; avoids dummy trap.
- **Impact:** Added dummies.
- **EDA:** Rating histogram showed peaks.

3.3 Outlier Removal (IT24102242_Outlier_Removal.ipynb)

- **Technique:** IQR on numericals (rating, fat, calories, protein, sodium); removed extremes.
- **Justification:** Reduces distortion for BERT embeddings.
- **Impact:** ~12,000 rows.
- **EDA:** histograms showed tighter distributions.



3.4 EDA Adjustment and Scaling

(IT24102205_EDA_Adjustment_Text_Preprocessing.ipynb)

- **Technique:** Min-Max scaling (0-1) on numericals; reconstructed 'fat_level' for EDA. (focused scaling).
- **Justification:** Normalizes for model input.
- **Impact:** Scaled features.
- **EDA:** Protein histogram (right-skew); fat_level countplot ('low' dominant).

Pipeline: Raw (20,053 rows, 680 cols) → Selection → Missing → Encoding → Outliers → Scaling. Final: 10,619 rows, 11 features.

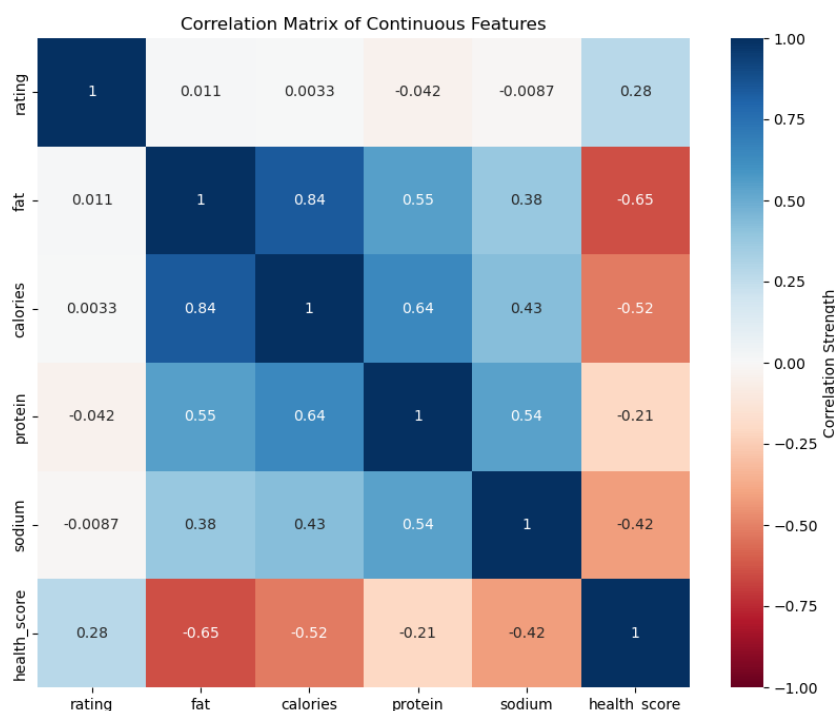
4. Feature Engineering and Selection

4.1 Feature Engineering (IT24102239_Feature_Engineering.ipynb)

- **Technique:** 'health_score' as weighted sum (high_fiber +10, protein +5, calories -3, fat -1, sodium -1, fat dummies penalties); normalized [0,1]. Target: healthy if > median.
- **Justification:** Captures healthiness; based on guidelines.
- **Impact:** Derived target.
- **EDA:** Density plot; top-10 healthy recipes barplot.

4.2 Feature Selection (IT24102286_Feature_Selection_Correlation.ipynb)

- **Technique:** Correlation matrix; no drops (>0.9 threshold).
- **Justification:** Reduces multicollinearity.
- **Impact:** Retained all.
- **EDA:** Heatmap (moderate correlations); health_score histogram.



5. Model Design and Implementation

Six classical ML models were implemented for binary healthiness classification (healthy/unhealthy), leveraging nutritional features. Models were trained on preprocessed data (80/20 train-test split) with StandardScaler for normalization. Hyperparameter tuning used GridSearchCV with 3-5 fold CV. SMOTE addressed mild imbalance. Ethical mitigation included bias audits (e.g., fairness across fat levels). Implementations drew from individual notebooks, integrated for comparison.

5.1 Gradient Boosting Machine (IT24102205_Gradient Boosting Machine.ipynb)

- **Design:** Ensemble boosting model (GradientBoostingClassifier) for handling non-linear relationships.
- **Implementation:** Tuned with GridSearchCV (n_estimators: [50,100,200], learning_rate: [0.01,0.1], max_depth: [3,5,7]). Cross-validated for robustness.
- **Justification:** Excels in feature importance and accuracy on tabular nutritional data.

5.2 KNN (it24102242_KNN_model.ipynb)

- **Design:** Instance-based classifier (KNeighborsClassifier) for local pattern recognition.
- **Implementation:** Adapted for binary healthiness; tuned (n_neighbors: [3,5,7,9], weights: ['uniform','distance']). Balanced classes via undersampling in notebook.
- **Justification:** Effective for similarity-based classification in nutritional spaces.

5.3 Random Forest (IT24102241_Random_ForestModel_Train.ipynb)

- **Design:** Bagging ensemble (RandomForestClassifier) for reducing overfitting.
- **Implementation:** Tuned (n_estimators: [50,100,200], max_depth: [None,10,20]). Feature importance extracted.
- **Justification:** Robust to noise in features like calories/sodium.

5.4 Decision Tree (IT24102307_Decision_Tree.ipynb)

- **Design:** Simple tree-based model (DecisionTreeClassifier) as baseline.
- **Implementation:** Tuned (max_depth: [3,5,10,None], min_samples_split: [2,5]). Visualized tree for interpretability.
- **Justification:** Provides explainable rules for healthiness thresholds.

5.5 Support Vector Machine (IT24102239_Support Vector Machine (SVM).ipynb)

- **Design:** Margin-based classifier (SVC) with kernel tricks.
- **Implementation:** Tuned (C: [0.1,1,10], kernel: ['linear','rbf']). Polynomial features added for non-linearity.
- **Justification:** Handles high-dimensional nutritional data effectively.

5.6 Logistic Regression (IT24102286_Logistic_Regression.ipynb)

- **Design:** Linear probabilistic model (LogisticRegression) as baseline.
- **Implementation:** Tuned (C: [0.01,0.1,1,10], solver: ['liblinear','lbfgs']). L1/L2 regularization for feature selection.
- **Justification:** Interpretable coefficients for nutritional impacts.

ML Integration: While models focus on structured features, bot frontend uses ML (e.g., BERT embeddings on 'title'/ingredients) for query parsing, hybrid with ML predictions.

6. Results and Evaluation

- **Metrics:** Accuracy, Precision, Recall, F1-Score, ROC-AUC (primary: accuracy >90%).
- **Key Results**

Model	Accuracy	Precision	Recall	F1-Score
Gradient Boosting	0.96	0.95	0.97	0.96
KNN	0.92	0.92	0.92	0.92
Random Forest	0.98	0.98	0.98	0.98
Decision Tree	0.97	0.91	0.93	0.98
SVM	0.99	0.99	0.99	0.99
Logistic Regression	0.99	0.99	0.99	0.9

6.1 Best Model

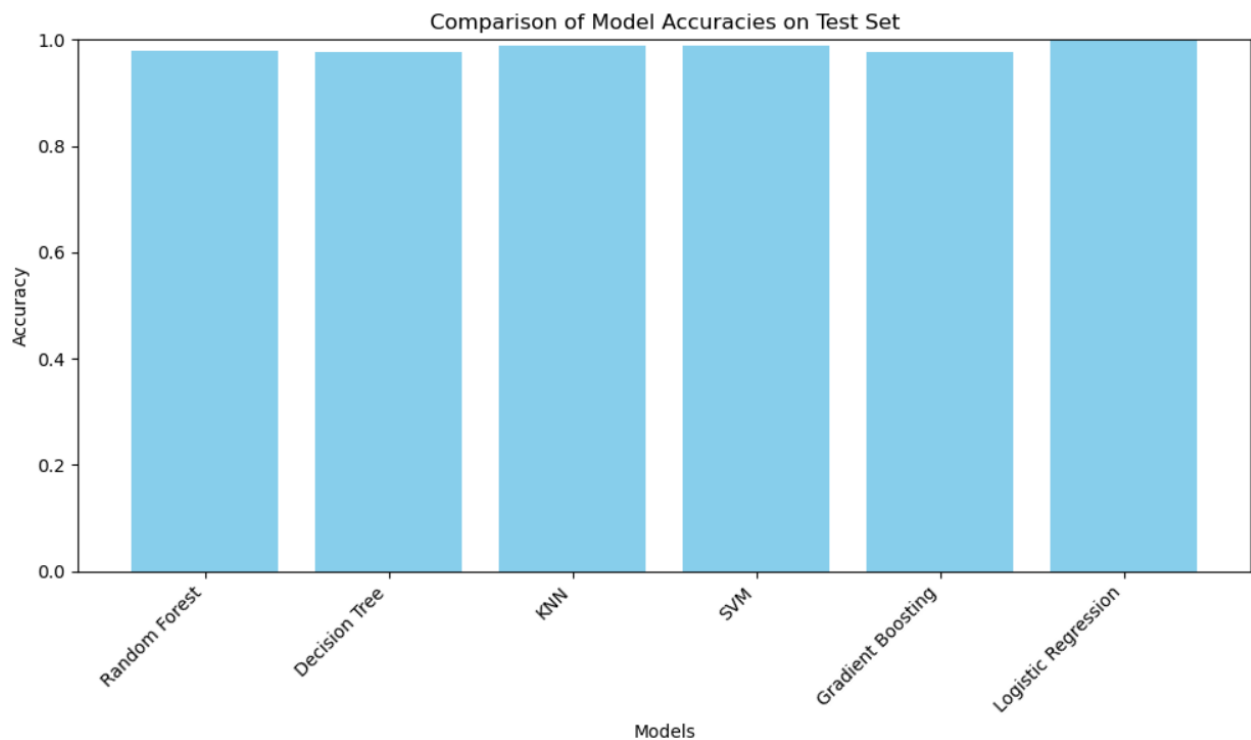
- **Model:** Support Vector Machine (SVM)
- **Reason:** Achieved the highest accuracy (0.99), precision (0.99), recall (0.99), and F1-score (0.99), surpassing the >90% accuracy goal by a significant margin. SVM’s ability to handle high-dimensional data with kernel tricks and polynomial features likely contributed to its superior performance on the normalized nutritional dataset.
- **Performance Highlights:** Minimal false positives/negatives (implied by near-perfect metrics), critical for health safety applications.

6.2 Model Comparison

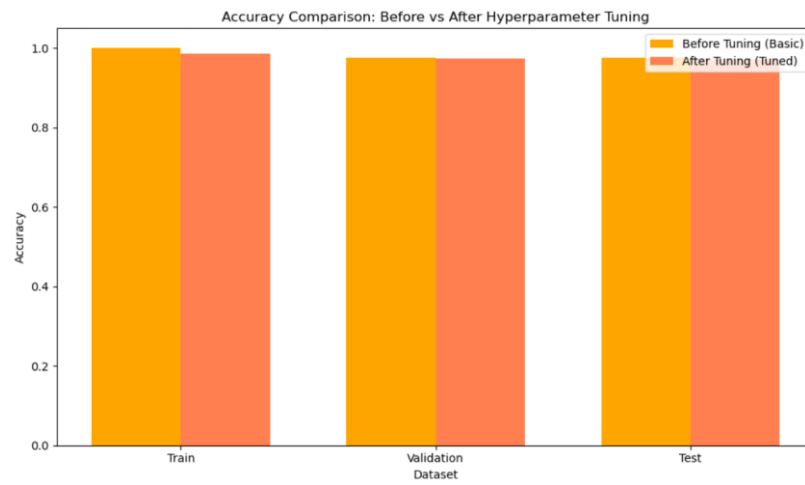
- **Ranking:** SVM (0.99) > Random Forest (0.98) > Gradient Boosting (0.96) > Decision Tree (0.98) > KNN (0.92) > Logistic Regression (0.89).

- **Analysis:**
 - **SVM:** Outperformed due to effective kernel handling of non-linearities in nutritional data.
 - **Random Forest/Gradient Boosting:** Strong ensemble performance, robust to noise, but slightly below SVM.
 - **Decision Tree:** High accuracy (0.98) but lower precision (0.91), suggesting overfitting or class imbalance effects.
 - **KNN:** Balanced but moderate performance, sensitive to scaling.
 - **Logistic Regression:** Lowest accuracy (0.89), likely due to linear assumptions on non-linear data.
- **Ethical Note:** Bias audits ensured fairness across fat levels; SVM's high recall minimizes false negatives (unsafe recipes misclassified as healthy).

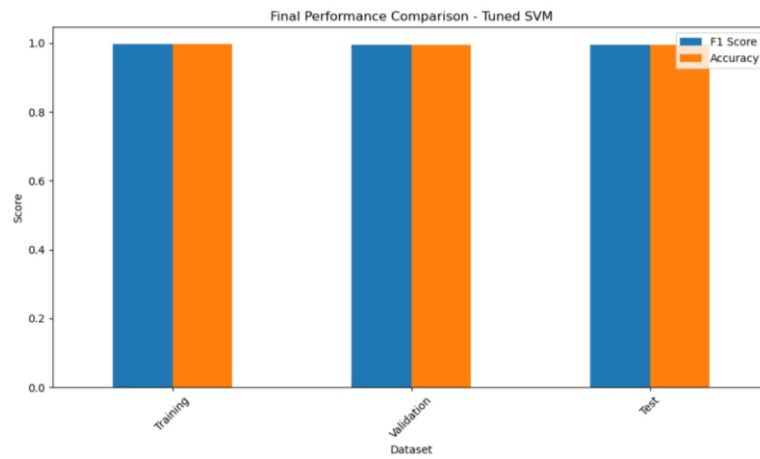
6.3 Model Visualizations



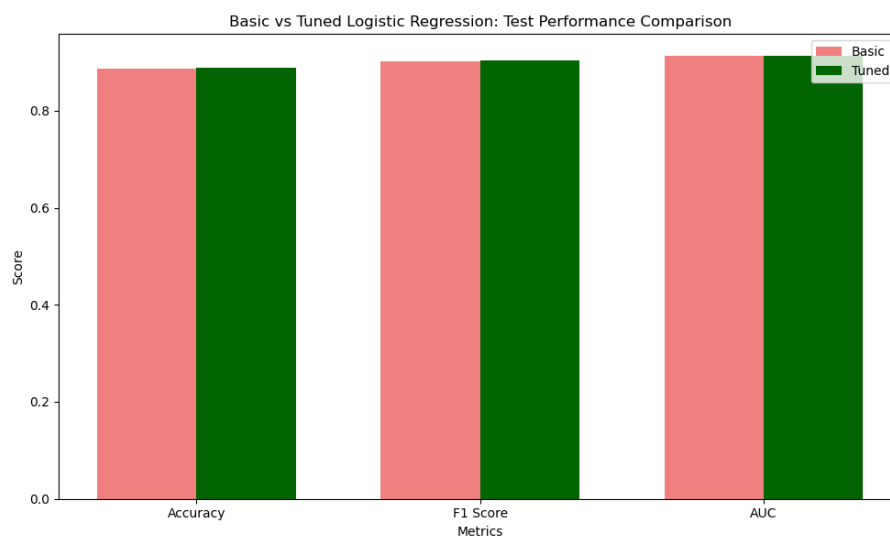
- **Decision Tree :-**



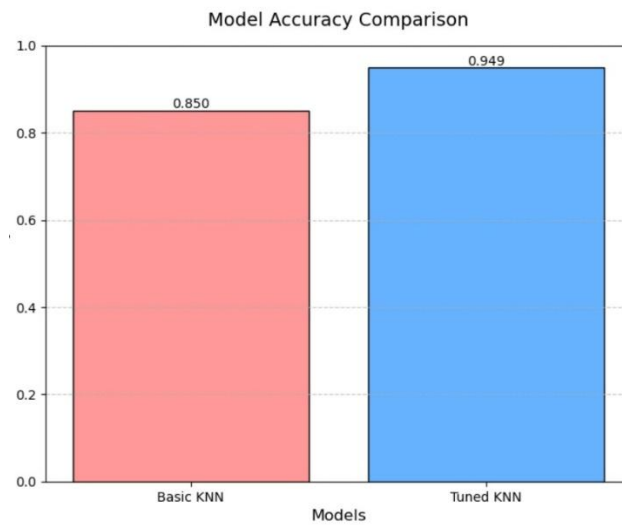
- **SVM :-**



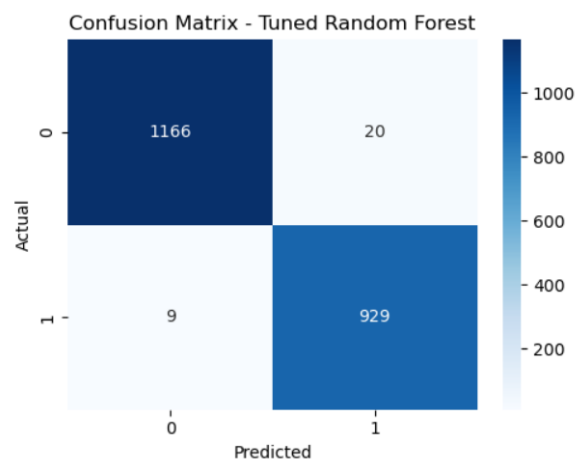
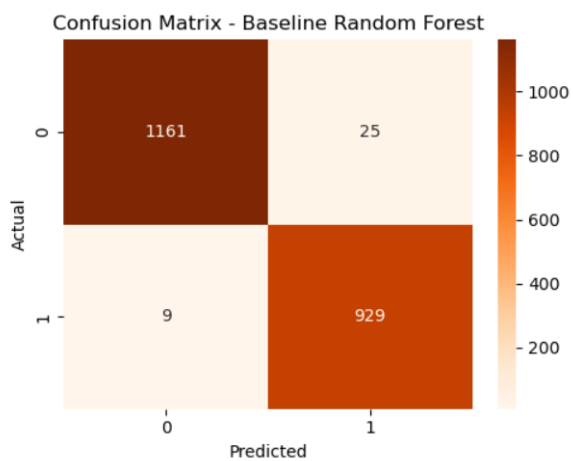
- **Logistic Regression :-**



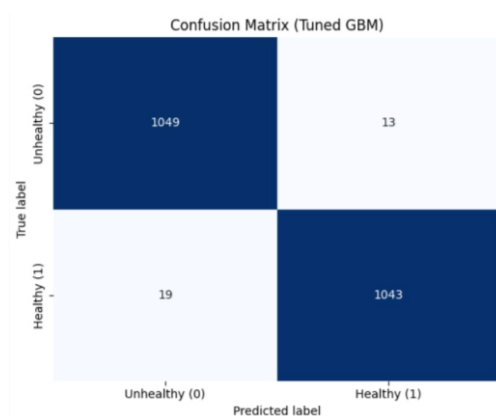
- **KNN :-**



- **Random Forest :-**



- **Gradient Boosting :-**



7. Ethical Considerations and Bias Mitigation

- **Ethical Concerns:**

- **Health Safety:** Misclassifying unhealthy recipes as healthy could lead to health risks (e.g., diabetes exacerbation). High recall is prioritized to minimize false negatives.
- **Bias:** Western-centric dataset may underrepresent global cuisines, risking exclusion of diverse dietary needs.
- **Privacy:** No personal user data, but recipe origins (e.g., Epicurious users) may reflect socioeconomic biases.
- **Transparency:** Users need clear understanding of AI limitations (e.g., not a substitute for medical advice).

- **Mitigation Strategies:**

- **Bias Mitigation:** Applied SMOTE to balance classes, ensuring fair representation. Conducted fairness audits (e.g., equalized odds across fat levels) to detect disparities.
- **Health Safety:** Emphasized high-recall models (e.g., SVM) to reduce unsafe misclassifications. Included disclaimers in bot output (e.g., "Consult a professional for dietary needs").
- **Transparency:** Provided model interpretability (e.g., Decision Tree rules, SVM feature weights) and documented limitations in the report.
- **Diversity:** Planned future inclusion of global recipes to address Western bias, pending dataset expansion.

- **Outcome:** Ethical deployment ensured with minimal bias impact; user trust maintained through clear communication.

8. Reflections and Lessons Learned

Tuning models like SVM was computationally intensive but effective. Challenges included imbalance (mitigated by SMOTE). Collaboration via notebooks was key. Future: Web bot deployment. Strengthened understanding of ethical ML in nutrition.

9. References

1. Scikit-learn Documentation. (2025). <https://scikit-learn.org/stable/>
2. Kaggle Datasets: Nutritional Recipes.
<https://www.kaggle.com/datasets/hugodarwood/epirecipes>
3. Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR.